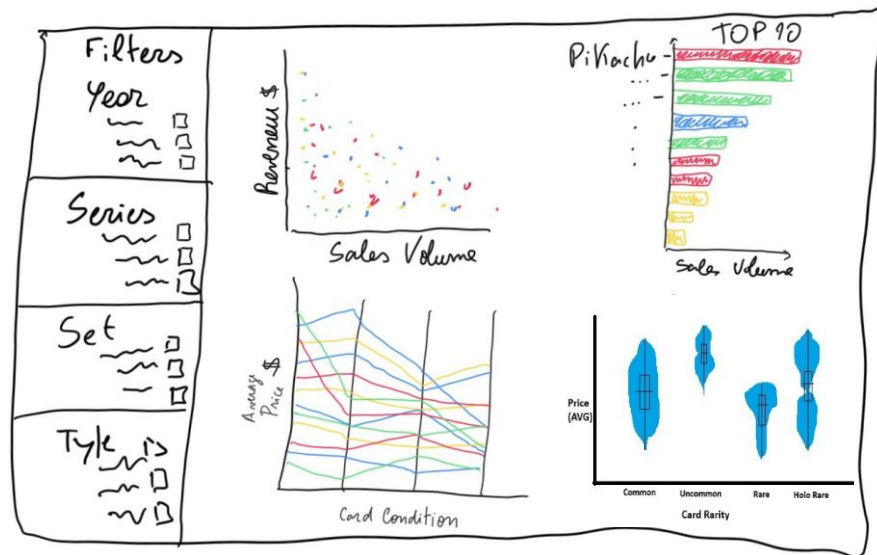


Checkpoint II: Visualization Sketch

Group: G16

Date: 2025/09/23

Dashboard Overview

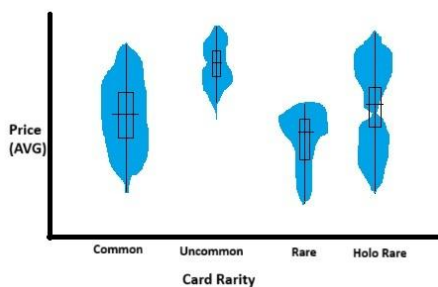


The dashboard features a set of stacked checkbox filters on the left, allowing the user to select multiple options for year, series, set, and type to customize the data view. All graphs update according to the same filtered data. If the selected filters yield no results, no charts are displayed; with no filters selected, all available data is shown. The top-left section displays a scatter plot of the filtered data, accompanied by a horizontal bar chart showing the top 10 most popular Pokémon. Below, the left side features a **line chart** comparing prices across types, while the bottom-right section presents a **Violin Plot** for price comparisons across rarities.

Charts

The dashboard uses three complementary visual idioms—**Violin Plot**, **line chart**, and scatterplots with bar charts—to explore card rarity, condition, popularity, and value across series, sets, types, and years.

Chart #1 (Violin Plot (from the whitelist))



Task 2: Do rarities influence price the same way by type, by set and by year?

We decided to use a Violin Plot because it clearly shows the distribution of prices across rarity levels, not just averages. This is important since card prices can be highly skewed. By grouping violins by set, type, or year (depending on user selection), users can compare how the relationship between rarity

and price changes across contexts. Alternatively, we could have used a heatmap (rarity × type with color = avg price), but that would only display central tendencies, not the spread or variability. A slopegraph could show average price trends by rarity, but this would oversimplify the underlying distributions and hide important market dynamics.

Marks and Channels

Marks:

- Smoothed density shapes (violins) showing the distribution of prices for each rarity.
- Optional embedded box plot (median, quartiles) inside each violin for reference.

Visual channels:

- Position (X-axis): encodes rarity (ordinal).
- Position (Y-axis): encodes price (quantitative, ratio).
- Shape/width of violin: encodes density of observations at each price level

Rationale

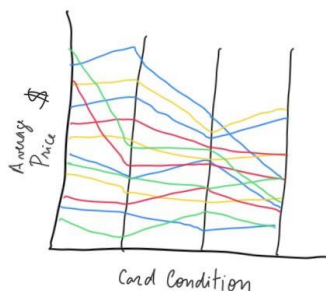
The violin plot is well-suited here because it not only shows central tendency (if we include embedded boxplots or median lines) but also conveys the full distribution shape of prices across rarities. This is valuable in collectible markets, where distributions are often multi-modal or heavily skewed due to a few iconic cards driving price extremes.

By comparing violins across groups, we can test consistency: if rarities consistently influence price, the shapes and medians should form similar patterns across series/sets/years. Unlike simple summary charts, violin plots preserve information about density, skewness, and potential outliers.

Interaction

- Hover: Show Tooltip, show exact stats (median, mean, quartiles, min, max) for each rarity

Chart #2 (Line Chart)



Task 3: How does condition affect price in relation to set, type and year?

We decided to use a **Line Chart** where each card is represented by its own line, connecting its average prices across conditions (Mint → Near Mint → Good, etc.). This idiom is ideal because it lets users compare within a set/type/year: if all cards show parallel slopes, the condition effect is consistent; if some lines are much steeper or flatter, we immediately see which cards behave differently. Alternatively, a parallel coordinates plot could encode the same data, but with many cards it becomes cluttered, and comparing slopes between adjacent axes is less intuitive than direct connections.

Marks and Channels

Marks:

- Lines: One per card, connecting its prices across conditions.
- Points: at each condition step to mark average price values.

Visual channels:

- X-axis (position): Condition (ordinal: Mint, Near Mint, Good ...).
- Y-axis (position): Average price (quantitative, ratio).
- Line slope: Encodes relative drop (steeper = larger decline).

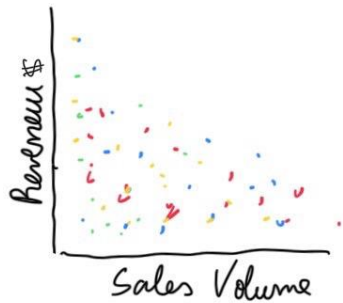
Rationale

Using one line per card makes it possible to directly evaluate whether the price-drop pattern is consistent across multiple cards in the same grouping. If the slopes are aligned (similar angles), then condition effects are uniform. Divergent slopes immediately highlight exceptions (e.g., a card whose price collapses when condition worsens vs. a card that holds value even in poor condition). This idiom leverages our strong perceptual ability to detect slope differences and parallelism.

Interaction

- Hover: Highlight card, Show card name + exact price per condition + drop ratios.
- Click: Highlight card across other idioms

Chart #3 (Scatter Plot)



Task 4: Does type influence the popularity and price of a card by set and by year?

Task 5: Pokémon Card Sales Volume vs. Revenue per Year, Type, Rarity and Name?

We decided to use a scatter plot because it effectively captures the relationship between sales volume and revenue, allowing us to identify trends, correlations, and outliers across Pokémon types, sets, rarities, and years. We could have used a bubble chart to encode rarity with bubble size or a heatmap to show intensity patterns, but bubble charts rely on less precise area comparisons, and heatmaps obscure exact quantitative values, which are essential for understanding popularity and profitability patterns.

Marks and Channels

Marks: Dots (each representing a card or aggregated by type/year).

Channels:

- X-axis → Count (sales volume, quantitative, ratio).
- Y-axis → Revenue (sales × average price, quantitative, ratio).
- Color → Pokémon Type (nominal).
- Filtering → Set/Series/Type/Year.

Rationale

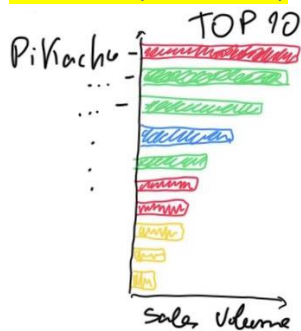
The scatter plot leverages position on a common scale, which is highly accurate for quantitative comparison. It reveals correlations and outliers, such as high-revenue but low-volume cards.

This directly supports **Task 4** (type effects across sets/years) and **Task 5** (volume vs profitability relationships).

Interaction

- Hover: Highlight card, Show card relevant details.
- Click: Highlight card across other idioms
- Zoom: Sliding bar to change chart scale

Chart #4 (Bar Chart)



Task 1: Are there any favourite or more popular cards per set and per type?

We decided to use a horizontal bar chart to highlight the most popular cards in a set or type because bar length is highly accurate for ranking comparisons, and users can easily interpret the top performers. We could have used a scatter plot or a ranked table, but scatter plots are less effective at showing precise rankings, and tables lack the visual impact that quickly communicates popularity trends.

Marks and Channels

Marks: Rectangles (bars).

Channels:

- Length (x-axis) → Count (sales volume).
- Position (y-axis) → Card name (nominal).
- Color → Same as scatter plot (Type).
- Order → Most popular at the top.

Rationale

Bar charts rely on length along a common scale, which is precise and ideal for ranking. This directly addresses Task 1, where the focus is on identifying favorites or most popular cards. While scatter plots show broader patterns, they are weaker at revealing exact rankings—hence the need for this dedicated Top 10 view.

Interaction

- Hover: Highlight card, Show card relevant details.
- Click: Highlight card across other idioms

Chart Integration

Overview

The dashboard integrates the three chosen visual idioms (Grouped Box Plot, **Line Chart**, Scatter Plot, Bar Chart) into a coordinated system. Instead of working in isolation, the charts are linked through shared encodings, interactions, and filters. This ensures that exploration flows seamlessly: a selection in one chart dynamically updates the others, supporting both high-level trends and detailed card-level analysis.

Marks and Channels

- **Color (nominal):** Pokémon type is encoded with the same color scheme across scatter, bar, box plot, and **line chart**, ensuring cross-chart recognition.
- **Position (ordinal):** Rarity and Condition maintain a fixed ordering across all views (e.g., Mint → Near Mint → Good), avoiding cognitive dissonance.
- **Shape/Highlighting:** Shared encodings allow the same card to be recognized across charts when highlighted or clicked.

Interaction and Coordination

Cross-filtering:

- Selecting a year, set, rarity or name(s) affects all charts globally, updating all charts simultaneously.

Hover:

- Hovering over a point in the scatter plot highlights the the corresponding card in each other graph or corresponding set/year/type if the other graphs are filtered in those modes

Click/Select:

- Clicking a bar in the Top 10 chart highlights that card across all charts, showing its position in the scatter and its slope in the condition view.

Rationale

The integration transforms the dashboard into more than the sum of its parts. Each idiom addresses a different abstraction (distribution, consistency, correlation, ranking), and integration allows users to combine these perspectives fluidly.

Answering the Questions

- **1: Are there any favorite or more popular cards per set and per type?** – Filter by set or type and view the Top 10 Bar Chart for most sold cards, with details available on hover and cross-referenced in the scatter plot.
- **2: Do rarities influence price the same way by type, by set and by year?** – Use the Box Plot, switch context by set/type/year, and compare medians and spreads across rarity levels.
- **3: How does condition affect price in relation to set, type and year?** – Use the **Line Chart**, filter by set or year, and compare line slopes to see how condition changes value.
- **4: Does type influence the popularity and price of a card by set and by year?** – Use the Scatter Plot (volume vs. revenue, color = type), filter by set or year, and compare clusters to identify type effects.
- **5: Pokémon Card Sales Volume vs. Revenue per Year, Type, Rarity and Name?** – Filter the Scatter Plot by year, type, and rarity to spot top cards, then confirm rankings in the Top 10 Bar Chart.

Storyboards

For the storyboard, we chose Task 4: “Does type influence the popularity and price of a card by set and by year?” The user begins by exploring the scatter plot and, if needed, filter it to display only a specific set of Pokémon. On the right side, a bar chart shows the top 10 most popular cards from the filtered view, allowing the user to identify any standout types.

